



Midterm Exam — Intermediate Algebra

Intermediate Algebra · Midterm Exam (Weeks 1–5) · numberbender.com

Name: _____

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Score: / 50

Learning Objectives

- Apply laws of exponents and factor polynomials (Week 1)
- Simplify and solve rational expressions and equations (Week 2)
- Simplify radicals, use rational exponents, solve radical equations (Week 3)
- Solve quadratic equations by factoring, completing the square, and the formula (Week 4)
- Solve and analyze quadratic inequalities (Week 5)

Choose the best answer for each item. This 50-item summative test covers Weeks 1–5. Show scratch work on separate paper where needed.

1. Simplify:

$$x^7 \cdot x^4$$

A.
 x^{28}

B.
 x^{74}

C.
 x^3

D.
 x^{11}

2. Simplify:

$$(x^4)^3$$

A.
 x^{43}

B.
 x^{64}

C.
 x^7

D.
 x^{12}

3. Simplify:

$$\frac{x^9}{x^4}$$

A.
 x^5

B.
 x^{13}

C.
 x^{36}

D.
 x^2

4. Simplify:

$$(2x^3)^4$$

A.
 $2x^{12}$

B.
 $8x^{12}$

C.
 $16x^{12}$

D.
 $16x^7$

5. Simplify:

$$(3x^2y^4)^2$$

A.

$$3x^4y^8$$

C.

$$6x^4y^8$$

B.

$$9x^4y^8$$

D.

$$9x^4y^6$$

6. Factor completely:

$$x^2 + 7x + 12$$

A.

$$(x + 3)(x + 4)$$

C.

$$(x + 1)(x + 12)$$

B.

$$(x - 3)(x - 4)$$

D.

$$(x + 2)(x + 6)$$

7. Factor completely:

$$x^2 - 16$$

A.

$$(x - 4)(x + 4)$$

C.

$$(x - 8)(x + 2)$$

B.

$$(x + 4)^2$$

D.

$$(x - 4)^2$$

8. Factor completely:

$$2x^2 + 7x + 3$$

A.

$$(x + 3)(x + 1)$$

C.

$$(2x - 1)(x - 3)$$

B.

$$(2x + 3)(x + 1)$$

D.

$$(2x + 1)(x + 3)$$

9. Factor completely:

$$3x^3 - 12x$$

A.

$$3x(x - 2)^2$$

C.

$$3x(x^2 + 4)$$

B.

$$3x(x - 2)(x + 2)$$

D.

$$3x(x - 4)(x + 4)$$

10. Factor completely (difference of cubes):

$$x^3 - 27$$

A.

$$(x - 3)^3$$

C.

$$(x - 3)(x^2 - 3x + 9)$$

B.

$$(x - 3)(x^2 + 3x + 9)$$

D.

$$(x + 3)(x^2 - 3x + 9)$$

11. Simplify:

$$\frac{x^2 - 9}{x + 3}$$

A.
 $x - 9$

C.
 $x + 3$

B.
 $x + 9$

D.
 $x - 3$

12. Simplify:

$$\frac{x^2 - 4}{x^2 - 4x + 4}$$

A.
 $\frac{x + 2}{x - 2}$

C.
 $\frac{x - 2}{x + 2}$

B.
 $x + 2$

D.
 $\frac{x - 2}{x - 2}$

13. Multiply:

$$\frac{x}{3} \cdot \frac{9}{x^2}$$

A.
 $\frac{3}{x}$

C.
 $\frac{x}{3}$

B.
 $3x$

D.
 $\frac{9}{x}$

14. Divide:

$$\frac{x^2}{2} \div \frac{x}{4}$$

A.
 $8x$

C.
 $\frac{x}{2}$

B.
 $\frac{x^2}{8}$

D.
 $2x$

15. Add:

$$\frac{1}{x} + \frac{2}{x}$$

A.
 $\frac{3}{2x}$

C.
 $\frac{2}{x^2}$

B.
 $\frac{3}{x^2}$

D.
 $\frac{3}{x}$

16. Add:

$$\frac{2}{x} + \frac{3}{x+1}$$

A.

$$\frac{5}{x(x+1)}$$

B.

$$\frac{5}{2x+1}$$

C.

$$\frac{5x+2}{x(x+1)}$$

D.

$$\frac{5x+2}{2x+1}$$

17. Subtract:

$$\frac{x}{x-1} - \frac{1}{x-1}$$

A.

$$1$$

B.

$$x-1$$

C.

$$0$$

D.

$$x$$

18. Simplify the complex fraction:

$$\frac{\frac{1}{x}}{\frac{1}{x^2}}$$

A.

$$\frac{1}{x^3}$$

B.

$$\frac{1}{x}$$

C.

$$x^2$$

D.

$$x$$

19. Simplify:

$$\frac{x^2 + 5x + 6}{x + 2}$$

A.

$$x-3$$

B.

$$x+3$$

C.

$$x+2$$

D.

$$x+6$$

20. Solve for x:

$$\frac{2}{x} = \frac{1}{3}$$

A.

$$x = \frac{2}{3}$$

B.

$$x = \frac{3}{2}$$

C.

$$x = 6$$

D.

$$x = \frac{1}{6}$$

21. Simplify:

$$\sqrt{50}$$

A.

$$10\sqrt{5}$$

C.

$$2\sqrt{5}$$

B.

$$25\sqrt{2}$$

D.

$$5\sqrt{2}$$

22. Simplify:

$$\sqrt{72}$$

A.

$$6\sqrt{2}$$

C.

$$36\sqrt{2}$$

B.

$$8\sqrt{2}$$

D.

$$6\sqrt{3}$$

23. Simplify:

$$\sqrt[3]{54}$$

A.

$$3\sqrt[3]{2}$$

C.

$$3\sqrt[3]{6}$$

B.

$$9\sqrt[3]{2}$$

D.

$$2\sqrt[3]{3}$$

24. Evaluate:

$$27^{\frac{1}{3}}$$

A.

$$3$$

C.

$$81$$

B.

$$9$$

D.

$$\frac{1}{3}$$

25. Evaluate:

$$16^{\frac{3}{4}}$$

A.

$$64$$

C.

$$8$$

B.

$$12$$

D.

$$4$$

26. Simplify (assume $x \geq 0$):

$$\sqrt{x^6}$$

A.
 x^2

C.
 x^3

B.
 x^9

D.
 x^{12}

27. Multiply and simplify:

$$\sqrt{3} \cdot \sqrt{12}$$

A.
 $6\sqrt{3}$

C.
36

B.
 $\sqrt{15}$

D.
6

28. Simplify:

$$\frac{\sqrt{18}}{\sqrt{2}}$$

A.
 $\sqrt{20}$

C.
3

B.
9

D.
4

29. Add:

$$2\sqrt{5} + 3\sqrt{5}$$

A.
 $5\sqrt{5}$

C.
 $5\sqrt{25}$

B.
 $6\sqrt{5}$

D.
 $5\sqrt{10}$

30. Solve for x:

$$\sqrt{x+3} = 5$$

A.
 $x = 8$

C.
 $x = 22$

B.
 $x = 28$

D.
 $x = 2$

31. Solve:

$$x^2 - 5x + 6 = 0$$

A.

$$x = -2, -3$$

C.

$$x = 1, 6$$

B.

$$x = 2, -3$$

D.

$$x = 2, 3$$

32. Solve:

$$x^2 - 9 = 0$$

A.

$$x = 3$$

C.

$$x = 9, -9$$

B.

$$x = \pm 9$$

D.

$$x = \pm 3$$

33. Solve:

$$x^2 + 6x + 9 = 0$$

A.

$$x = \pm 3$$

C.

$$x = -9$$

B.

$$x = -3$$

D.

$$x = 3$$

34. Solve:

$$x^2 = 49$$

A.

$$x = \pm 7$$

C.

$$x = \pm 49$$

B.

$$x = 24.5$$

D.

$$x = 7$$

35. Solve:

$$x^2 - 2x - 15 = 0$$

A.

$$x = 5, -3$$

C.

$$x = -5, 3$$

B.

$$x = 15, -1$$

D.

$$x = 5, 3$$

36. Solve using the quadratic formula:

$$x^2 + 4x + 1 = 0$$

A.

$$x = -4 \pm \sqrt{3}$$

C.

$$x = 2 \pm \sqrt{3}$$

B.

$$x = -2 \pm \sqrt{5}$$

D.

$$x = -2 \pm \sqrt{3}$$

37. Solve:

$$2x^2 - 7x + 3 = 0$$

A.

$$x = \frac{1}{2}, -3$$

B.

$$x = -3, -\frac{1}{2}$$

C.

$$x = 3, 2$$

D.

$$x = 3, \frac{1}{2}$$

38. Solve:

$$x^2 + x - 6 = 0$$

A.

$$x = 6, -1$$

B.

$$x = -2, 3$$

C.

$$x = 2, 3$$

D.

$$x = 2, -3$$

39. Find the discriminant:

$$x^2 - 4x + 5 = 0$$

A.

$$-36$$

B.

$$-4$$

C.

$$36$$

D.

$$4$$

40. Solve:

$$x^2 - 6x + 8 = 0$$

A.

$$x = 1, 8$$

B.

$$x = 2, -4$$

C.

$$x = 2, 4$$

D.

$$x = -2, -4$$

41. Solve:

$$x^2 - 5x + 6 < 0$$

A.

$$x < 2 \text{ or } x > 3$$

B.

$$(-3, -2)$$

C.

$$[2, 3]$$

D.

$$(2, 3)$$

42. Solve:

$$x^2 - 9 > 0$$

A.

$$x < -3 \text{ or } x > 3$$

B.

$$x > 3$$

C.

$$-3 < x < 3$$

D.

$$(-3, 3)$$

43. Solve:

$$x^2 - x - 12 \leq 0$$

A.

$$[-3, 4]$$

C.

$$x \leq -3 \text{ or } x \geq 4$$

B.

$$(-3, 4)$$

D.

$$[-4, 3]$$

44. Solve:

$$x^2 + 2x - 8 \geq 0$$

A.

$$[-4, 2]$$

C.

$$x \leq -4 \text{ or } x \geq 2$$

B.

$$x \leq -2 \text{ or } x \geq 4$$

D.

$$-4 \leq x \leq 2$$

45. Solve:

$$x^2 - 4 < 0$$

A.

$$(2, \infty)$$

C.

$$x < -2 \text{ or } x > 2$$

B.

$$(-2, 2)$$

D.

$$[-2, 2]$$

46. Solve:

$$x^2 - 4x \leq 0$$

A.

$$[-4, 0]$$

C.

$$(0, 4)$$

B.

$$[0, 4]$$

D.

$$x \leq 0 \text{ or } x \geq 4$$

47. Solve:

$$x^2 + 5x + 6 > 0$$

A.

$$(-3, -2)$$

C.

$$x > -2$$

B.

$$x < -3 \text{ or } x > -2$$

D.

$$-3 < x < -2$$

48. Solve:

$$x^2 \leq 0$$

A.

$$x \geq 0$$

C.

no solution

B.

$$x = 0 \text{ only}$$

D.

all reals

49. Solve:

$$x^2 + 1 > 0$$

- A.
no solution
- C.
 $x > -1$

- B.
all real numbers
- D.
 $x \neq 0$
-

50. Solve:

$$x^2 - 2x - 3 > 0$$

- A.
 $-1 < x < 3$
- C.
 $x < 1$ or $x > 3$

- B.
 $[-1, 3]$
- D.
 $x < -1$ or $x > 3$
-



Answer key below. Items 1–10 Exponents & Factoring · 11–20 Rational Expressions · 21–30 Radicals · 31–40 Quadratic Equations · 41–50 Quadratic Inequalities.

Answer Key

1. D	2. D	3. A	4. C	5. B	6. A	7. A	8. D	9. B	10. B
11. D	12. A	13. A	14. D	15. D	16. C	17. A	18. D	19. B	20. C
21. D	22. A	23. A	24. A	25. C	26. C	27. D	28. C	29. A	30. C
31. D	32. D	33. B	34. A	35. A	36. D	37. D	38. D	39. B	40. C
41. D	42. A	43. A	44. C	45. B	46. B	47. B	48. B	49. B	50. D